Electronic and Telecommunication Engineering

Section 5

Medical Ultrasound

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Acknowledgement: Dr. Nuwan Dayananda for the content

BM3122- Medical Imaging
2025/ 09/ 18

Medical Ultrasound

- Least expensive and most portable clinical imaging technique
- Morphological and structural information as well as real-time blood flow measurements
- Ultrasound is the second most used imaging modality
  - Behind projection x-ray

Medical Ultrasound cont.

- Advantages
  - High frame rate (high temporal resolution)
    - 100's of 2D frames per second is possible
  - Flow velocity measurements using Doppler US
    - Functional information on anatomical information
  - Inexpensive equipment
    - ~1/5 price of a CT scanner
    - <1/10 price of an MRI scanner
  - Portable
    - Valuable in ER and ICU settings
    - Well suited for image-guided interventional procedures

Medical Ultrasound cont.

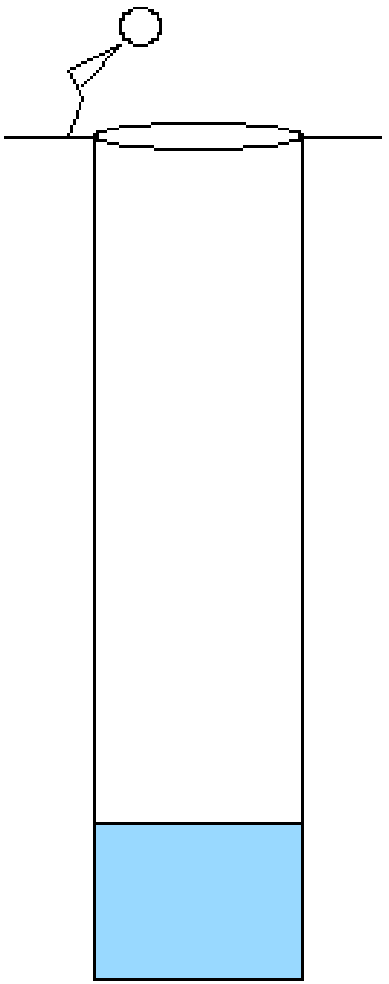
- Disadvantages
  - Limited field of view
    - Compared to CT, MRI
  - Images are operator dependent
  - Images are difficult to interpret
  - Inappropriate for examining bony or air-filled structures

Clinical Applications

- Obstetrics (~30%)
  - No ionizing radiation or strong magnetic fields
- Cardiovascular (~30%)
  - Many applications including detecting mitral valve and septal insufficiencies
- Abdominal (~30%)
  - Nearly all abdominal organs
- Other (~10%)
  - Includes: breast, eye, skin, etc.
  - Obstructive atherosclerosis in the carotids

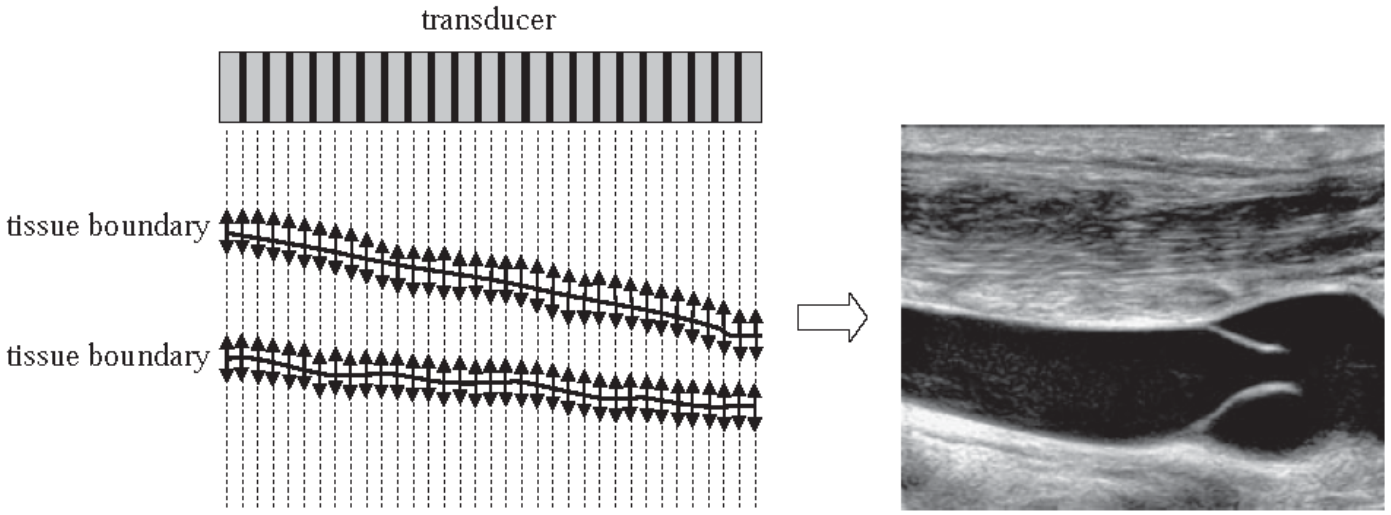
Ultrasound Principle

- When you shout into a well
  - The sound is reflected (echoes) off the surface
  - If the time for the echo to return is measured and if the speed of sound is known,
    - the depth of the well can be calculated fairly accurately
- The same principle is used in medical ultrasound imaging



Ultrasound Principle cont.

- A transducer sends a series of pressure waves through the tissue
- A small fraction of the energy is backscattered towards the transducer at boundaries
- Strength and time difference of the signal is used to produce the ultrasound image

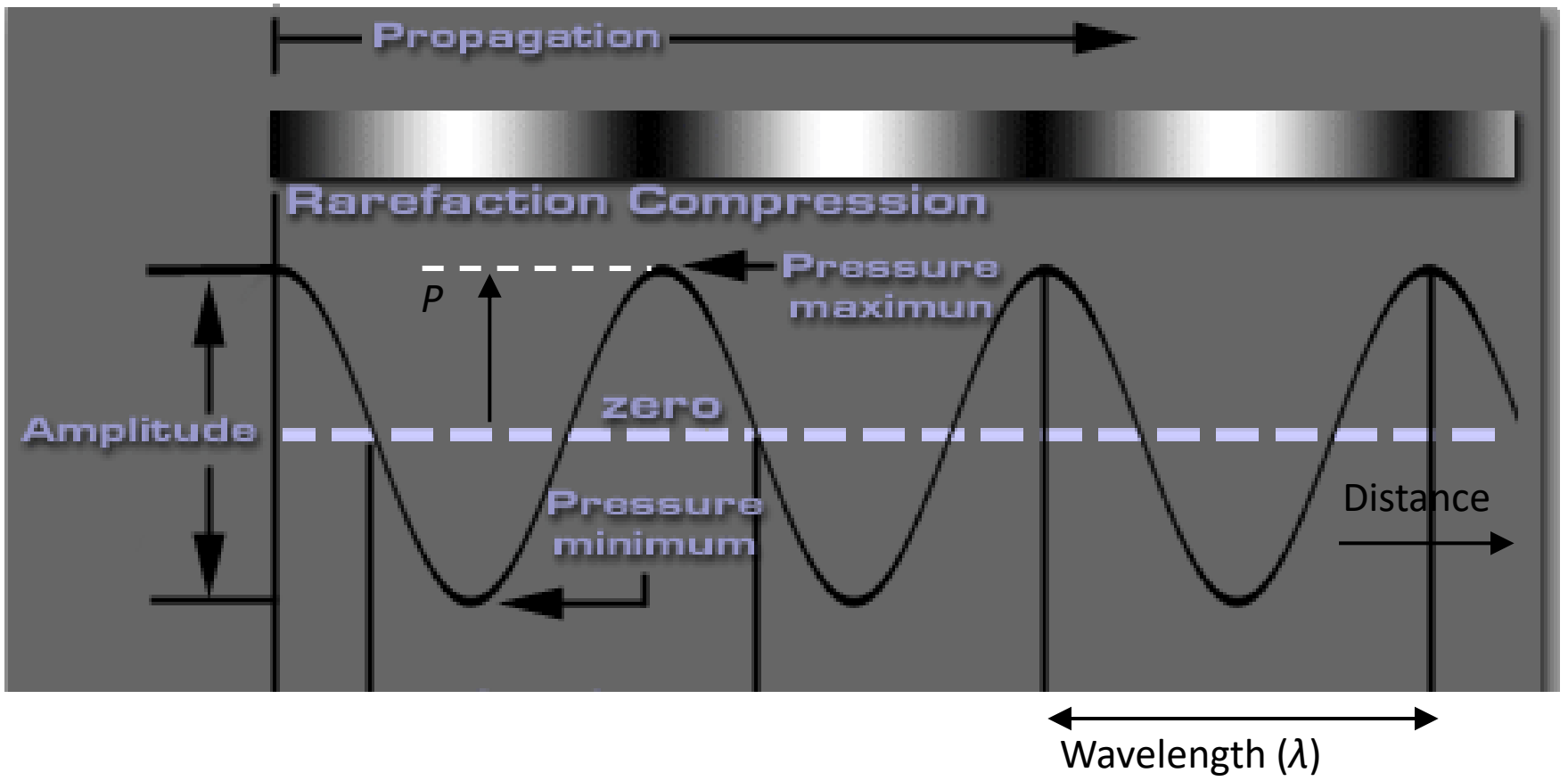


Ultrasound Principle cont.

- Ultrasound
  - Frequency greater than 20kHz
    - Above the audible range
- Medical ultrasound
  - Frequency in the range **2-15 MHz**
  - Echoes of sound that is converted to an image

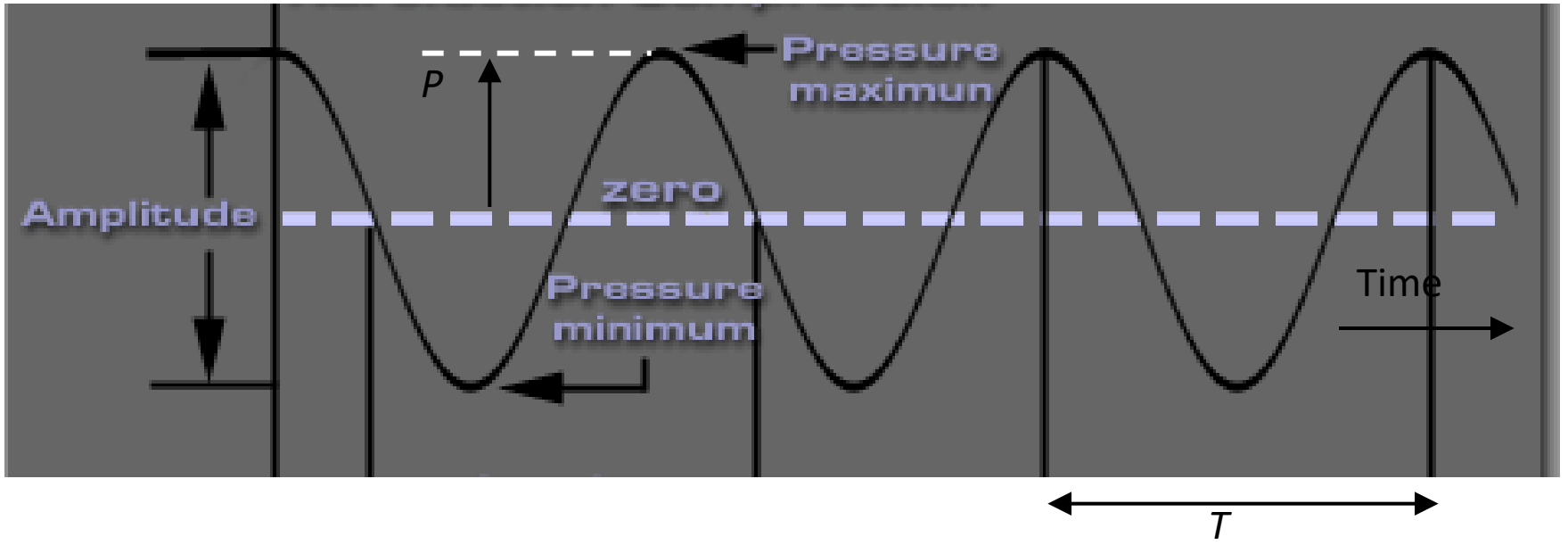
Ultrasound Wave

- Snapshot of a frozen wave



Ultrasound Wave cont.

- Measurement of wave at a fixed position



Ultrasound Wave cont.

- The speed (c) of the ultrasound wave through tissue is determined by:
  - Tissue density (ρ) and
  - Compressibility (κ)

$$c = \frac{1}{\sqrt{\kappa\rho}}$$

- Characteristic acoustic impedance (Z) of tissue:

$$Z = \rho c = \rho \frac{1}{\sqrt{\kappa\rho}} = \sqrt{\frac{\rho}{\kappa}}$$

Ultrasound Wave cont.

- Acoustic properties of biological tissues

	$Z \times 10^5$ (g cm^{-2} s^{-1})	Speed of sound (m s^{-1})	Density (gm^{-3})	Compressibility x10^{11} (cm g^{-1} s^2)
Air	0.00043	330	1.3	70 000
Blood	1.59	1570	1060	4.0
Bone	7.8	4000	1908	0.3
Fat	1.38	1450	925	5.0
Brain	1.58	1540	1025	4.2
Muscle	1.7	1590	1075	3.7
Liver	1.65	1570	1050	3.9
Kidney	1.62	1560	1040	4.0

Acoustic Impedance

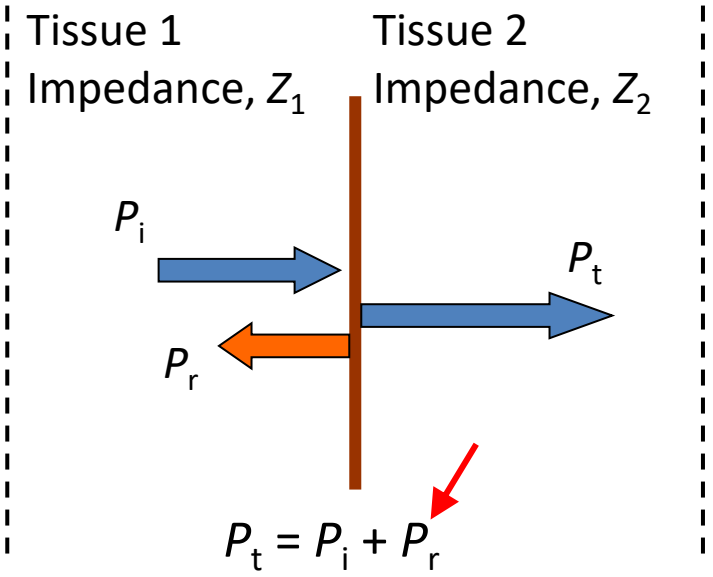
- Ultrasound images display variations in the acoustic impedance of tissue
  - Variations in density and compressibility, or
  - Variations in density and sound speed
- Bone and air-filled structures
  - Strong reflections
- Soft tissue boundaries
  - Weaker reflections
  - Important in medical imaging to get the depth information

Ultrasound-Tissue Interactions

- Reflection
  - Specular
    - Mirror-like reflection
  - Scattering
    - From small targets
- Attenuation
  - Absorption
    - Viscous reactions that convert mechanical to thermal energy
  - Redirection
    - Sound energy deflecting outside imaging beam

Ultrasound-Tissue Interactions cont.

- At perpendicular incidence



Reflection coefficient

$$R = \frac{P_r}{P_i} = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

Transmission coefficient

$$T = \frac{P_t}{P_i} = 1 + R = \frac{2Z_2}{Z_1 + Z_2}$$

P_i = amplitude of incidence wave
P_t = amplitude of transmitted wave
P_r = amplitude of reflected wave
Z_1 = acoustic impedance of tissue 1
Z_2 = acoustic impedance of tissue 2

Attenuation (Absorption)

- Sound intensity gain *I(d)* (in dB w.r.t. incident) at round trip distance, *d* (cm)

$$I(d) = -\mu f d$$

Where μ = attenuation coefficient in *dB/(cm.MHz*)

f = frequency (*MHz*)

- In most soft tissue
  - $0.5 \leq \mu \leq 1.0$ *dB/(cm.MHz*)
- In air
  - $\mu = 45$ *dB/(cm.MHz)*
- In bone
  - $\mu = 8.7$ *dB/(cm.MHz)*

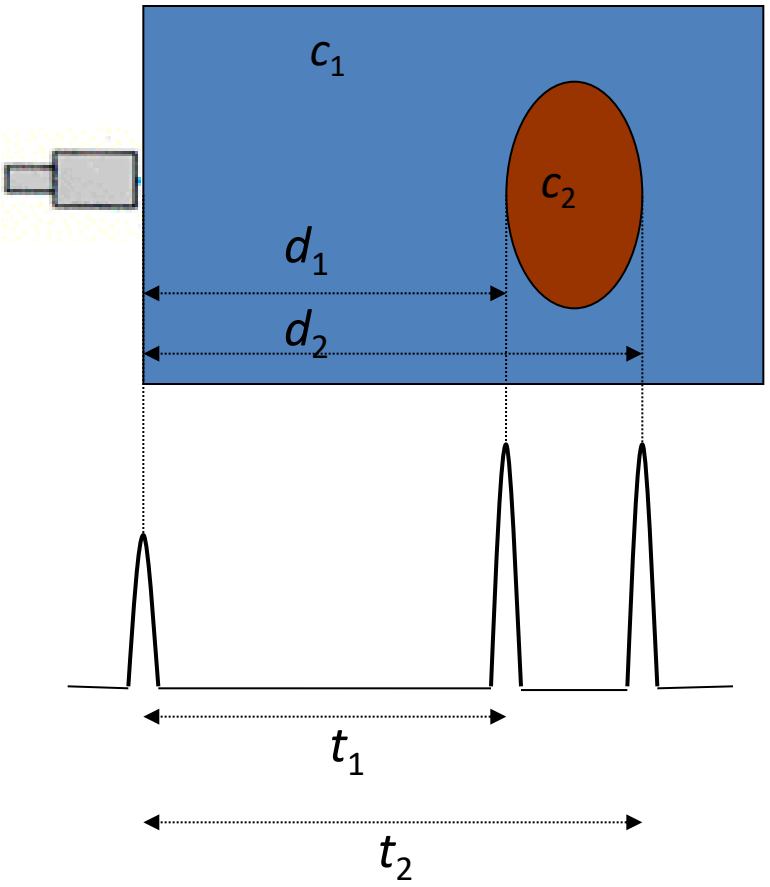
A-mode Imaging

- A-mode = “Amplitude” mode

$$d_1 = \frac{c_1 t_1}{2}$$

Round Trip

$$d_2 = d_1 + \frac{c_2 (t_2 - t_1)}{2}$$



2 equations and 4 unknowns

So, scanners assume,

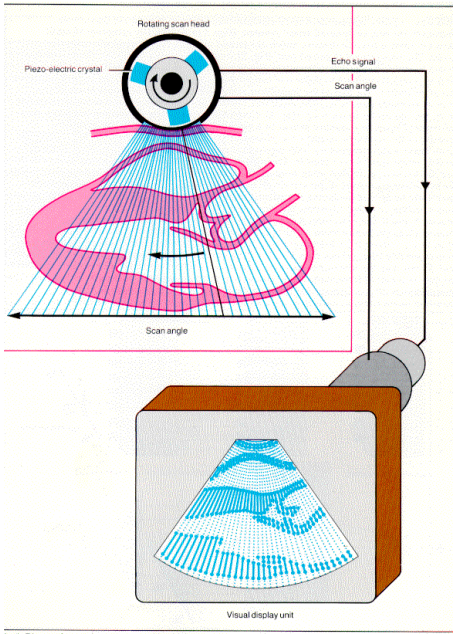
$c_1 = c_2 = 1540$ m/s

A-mode Imaging cont.

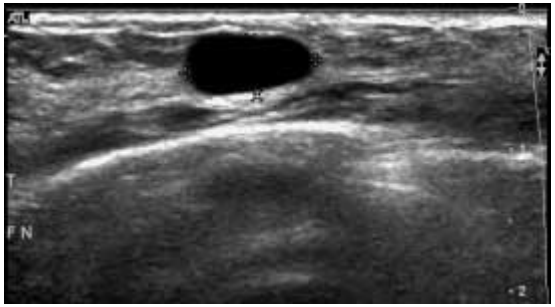
- Envelop detection
  - Magnitude (envelope) of echoes displayed with calculated distances to the reflected surface
- Simplest approach
  - Full wave rectification followed by low-pass filtering
- Modern scanners employ more accurate digital signal processing methods

B-Mode Imaging

- B-Mode
  - "Brightness" mode
  - Strength of the echoes at different locations
  - Distance calculated from time differences



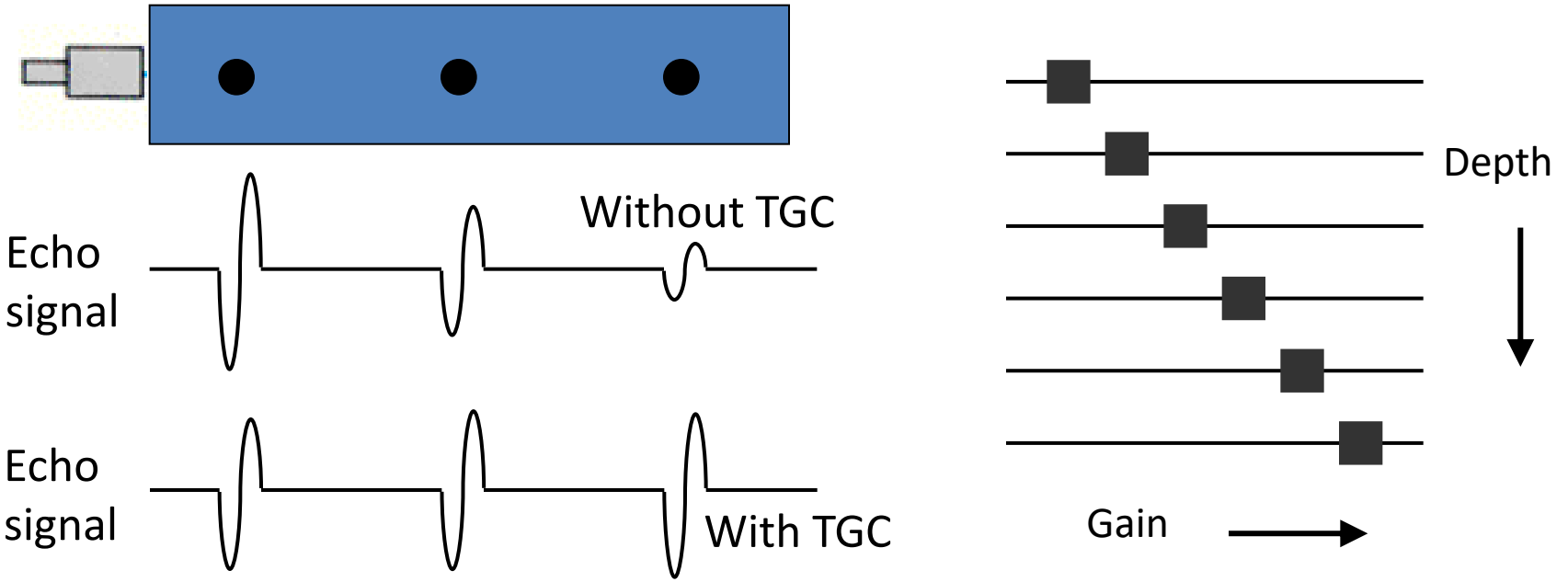




Time Gain Compensation (TGC)

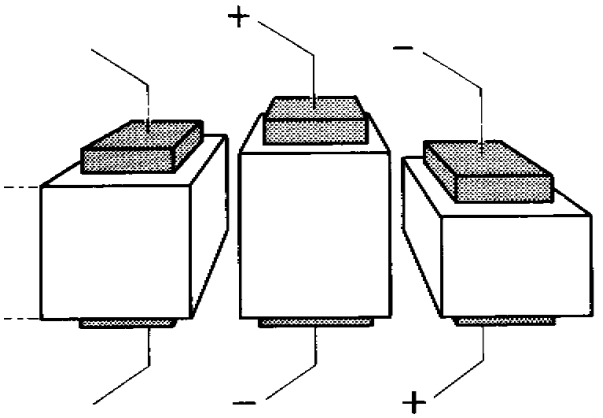
- TGC slope
  - Set by the operator to compensate for attenuation
  - Slider controls on scanner console

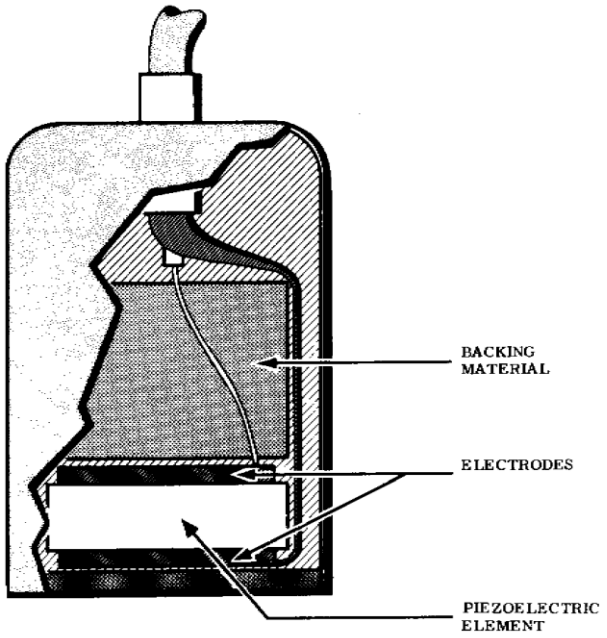
With TGC, $\boldsymbol{I(d) = -\mu f d + (\text{TGC slope}) \times d}$



Piezoelectric Transduction

- Piezoelectric crystal
  - Pressure generated when electric field is applied
  - Electric energy generated when pressure is applied
- Alternating voltage
  - Cyclic expansion and compression
- Clinical transducers
  - Lead Zirconate Titanate (PZT)
  - Composite of PZT and epoxy





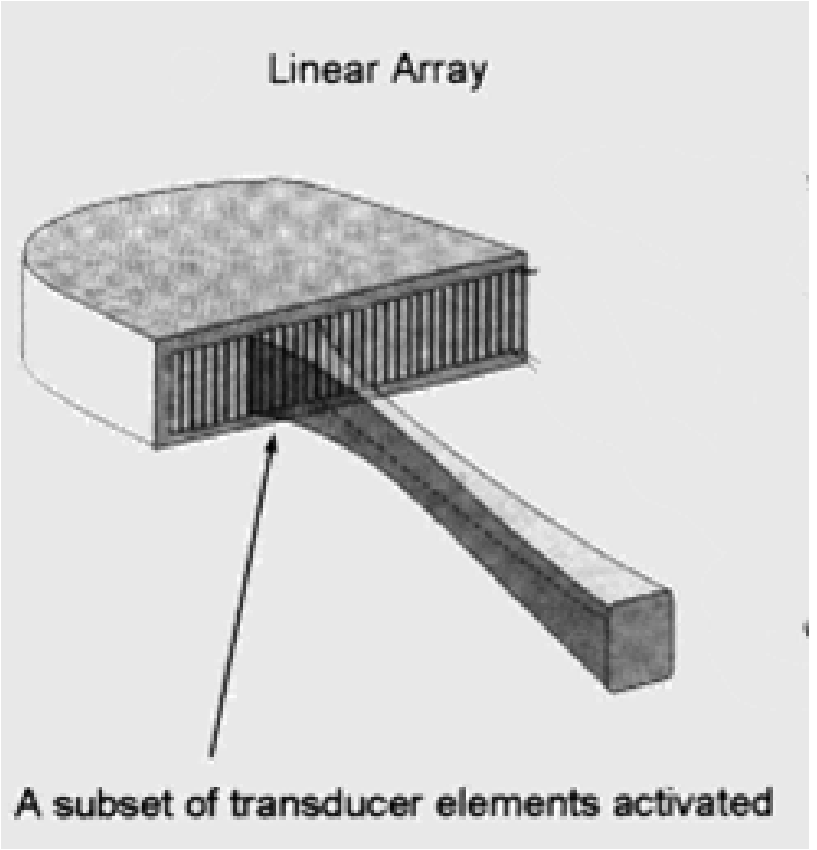
Mechanical Beamsteering

- Beamsteering
  - To acquire multiple lines

- Mechanical beamsteering
  - Rotation of piezoelectric element in the probe
  - Circular sector image
  - Rarely used in modern clinical scanners

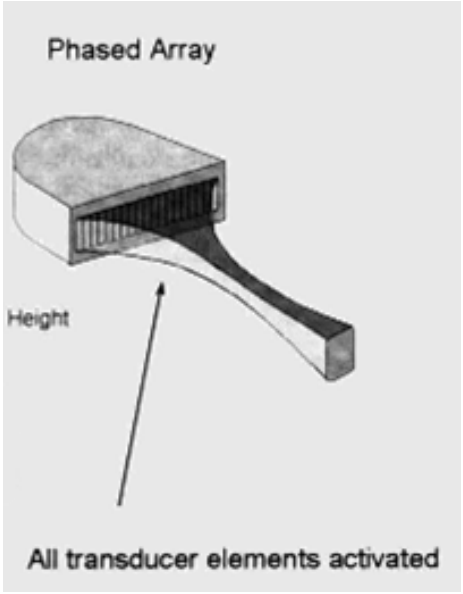
Electronic Beamsteering

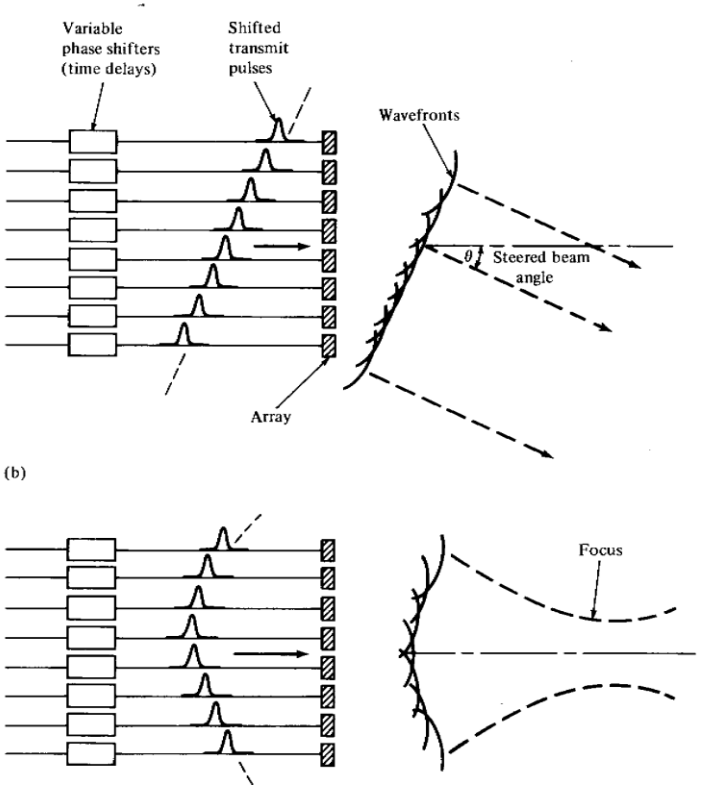
- Linear array
  - Subset of elements used in each scan
  - Usually rectangular image
- Curvilinear array
    - For sector image



Electronic Beamsteering cont.

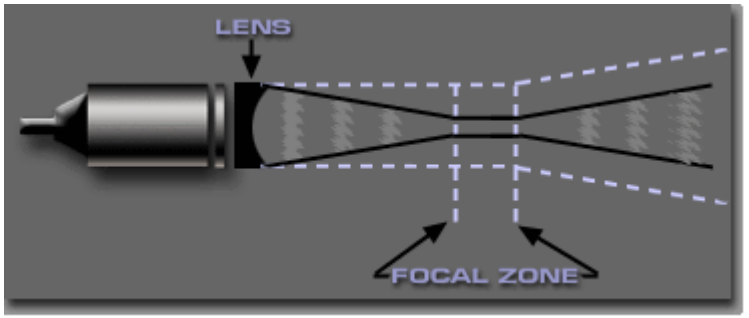
- Phased array
  - All elements used in each scan
  - Sector image
  - Smaller than linear arrays

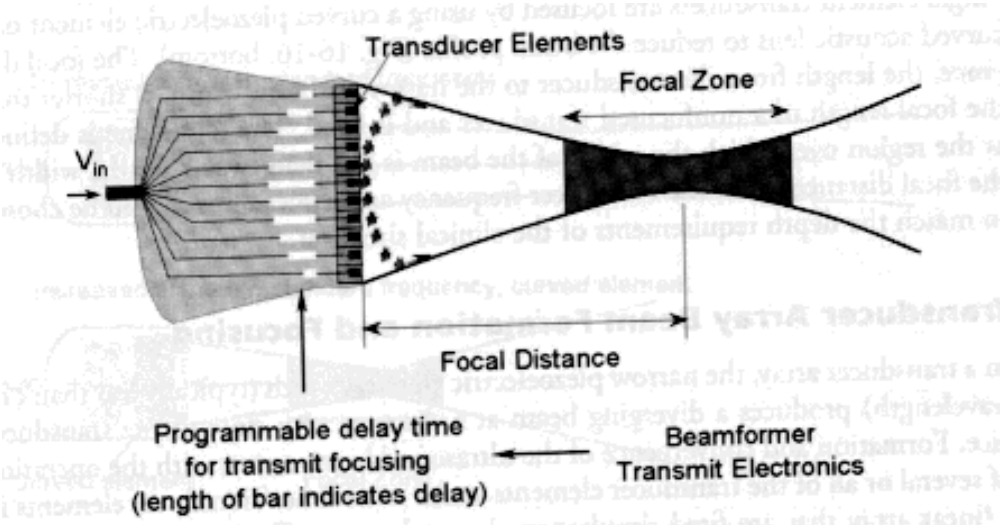




Focused Beam

- To achieve an acceptable lateral resolution
- Mechanical focusing
  - Used with mechanical scanners
- Electronic focusing
  - Used with arrays





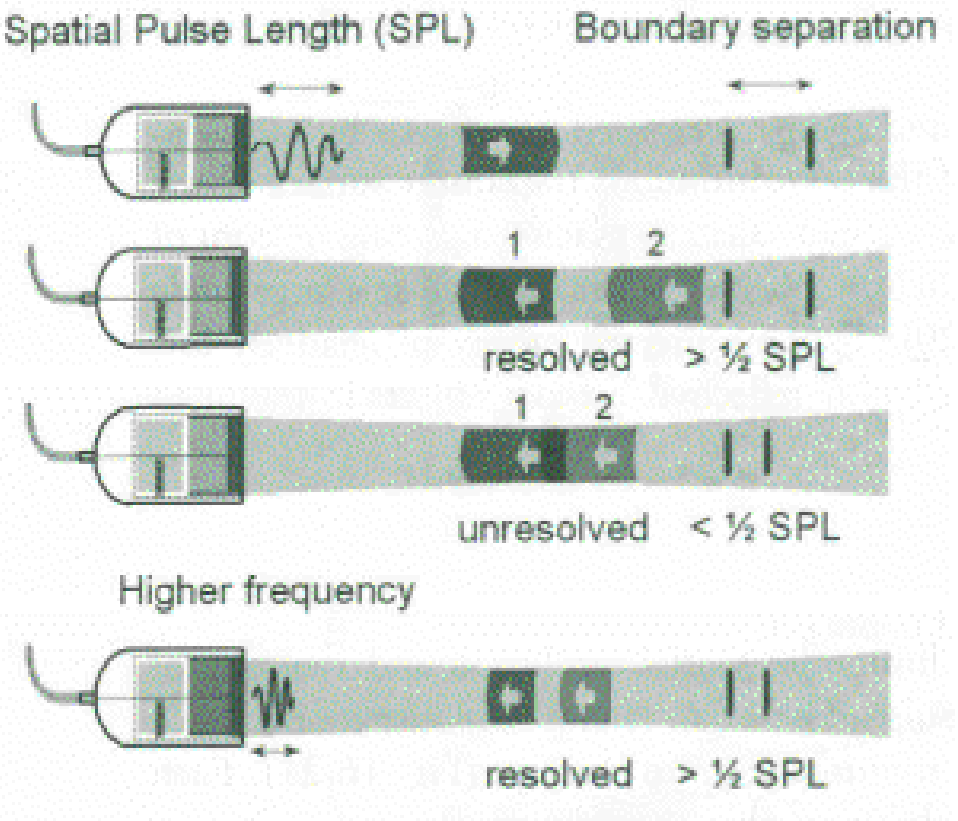
F-number = *F/L*

Axial Resolution

- Ability to discern two closely spaced objects in the direction of the beam

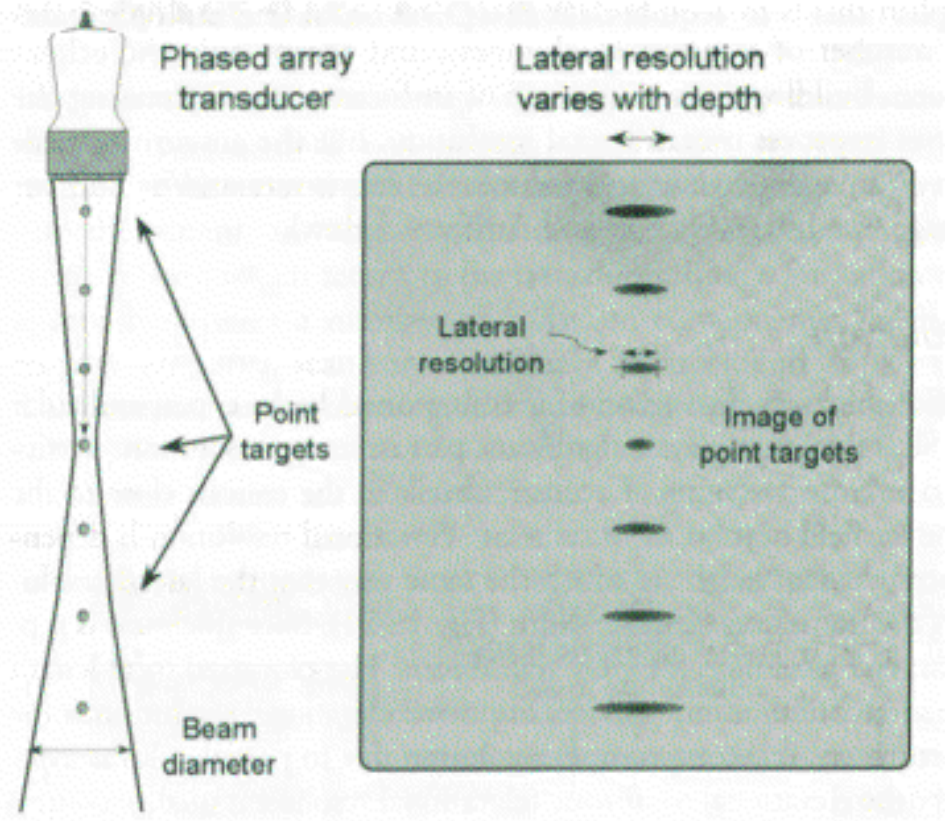
  Axial resolution =

  0.5 x (pulse width)



Lateral Resolution

- Ability to discern as separate two closely spaced objects perpendicular to the beam direction



Resolution Vs Penetration

$$\text{Axial Resolution} = 0.5 \times \text{Pulse Length} = \frac{N_c \lambda}{2} = \frac{N_c c}{2f}$$

N_c = number of cycles per pulse

$$\text{Lateral Resolution at Focus} = \text{Wavelength} \times \text{F}-\text{number} = \lambda \frac{F}{L} = \frac{cF}{fL}$$

$$\text{Attenuation} = \mu f d$$

Increasing frequency improves spatial resolution but reduces penetration depth.

Resolution Vs Penetration cont.

- For addressing trade-off
  - Use high frequencies for superficial anatomy (e.g. Eye, peripheral vessels), and lower frequencies for large/deep organs (e.g. heart, liver)
  - Position the transducer closer to the anatomy
    - Trans-oesophageal or intra-vascular probes
  - Harmonic imaging
    - Transmit at *f*, and receive at 2*f* or 3*f* (only travels the return path)

Multi-dimensional Arrays

- Allows focusing in 3D
- 1.5-dimensional array
  - A small number of rows is added, typically 3 – 10
  - Limited focusing on the elevation direction
  - Requires mechanical movement in the third dimension for 3D data acquisition



Multi-dimensional Arrays cont.

- 2-dimensional array
  - A large number of rows is added
  - Up to a value equal to the number of elements in each row
  - Full 3D data acquisition





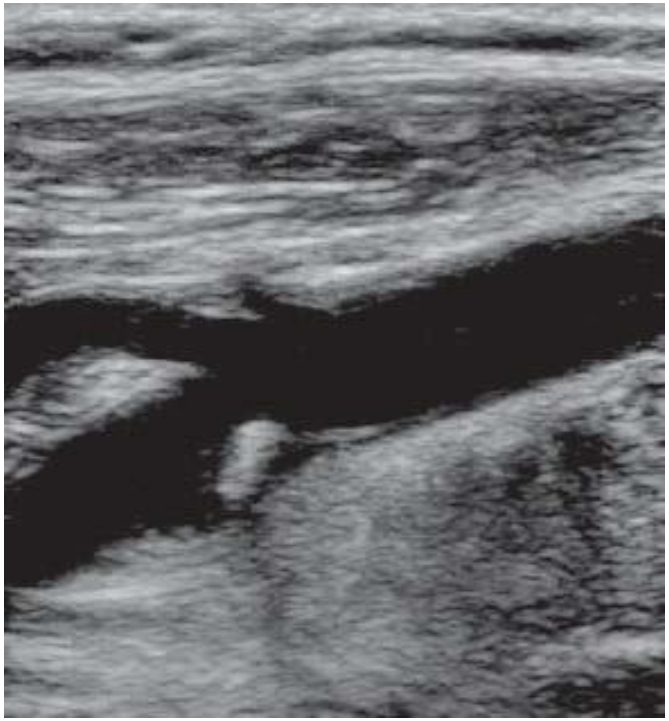
Compound Scanning

- Mainly to reduce the degree of speckle in the image
- Acquires an ultrasound image from multiple angles and then combines the images together
  - Using a phased array with different beam forming schemes
  - Separate angles have different contributions from speckle, but the same ‘true’ features are present in all the views
  - Combination of the images produces a composite image with reduced speckles
  - Added advantage: any boundary or structure parallel to the beam will also be visible

Compound Scanning cont.

- Example: compound imaging with different orientations
  - Left: a conventional B-mode scan
  - Right: a compound scan with 9 different orientations





Compound Scanning cont.

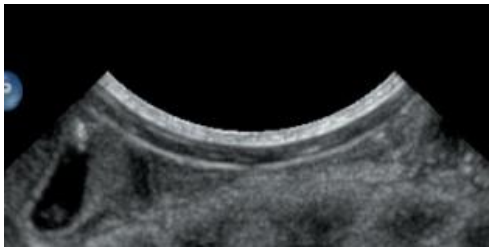
- Frequency compounding
  - Multiple frequencies are used to construct the composite image
  - The appearance of the speckle pattern depends upon the frequency, and so does not add constructively between views
  - The transducer must have a large enough bandwidth to cover the frequency range

Artefacts in Ultrasound Imaging

- Must be recognized to avoid incorrect image interpretation
  - Once recognized it can give useful diagnostic information
- Speckle
  - Arises from the constructive and destructive interference of ultrasound scattered from very small structures within tissue
  - Gives rise to high and low intensities not correlated directly with any particular structure

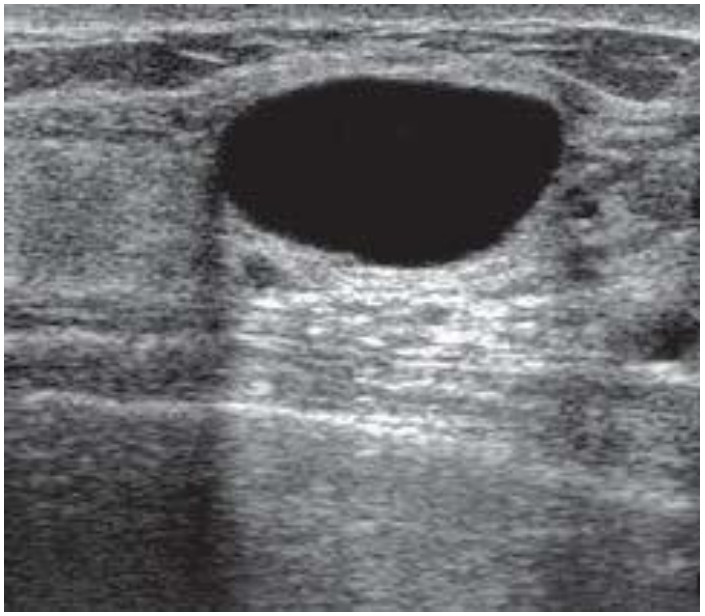
Artifacts in Ultrasound Imaging cont.

- Reverberations
  - Occur if there is a very strong reflector close to the transducer surface
  - Multiple reflections occur between the surface of the transducer and the reflector
  - Reflections appear as a series of repeating lines in the image
  - Relatively simple to detect due to the equidistant nature of the lines
  - Occur when the ultrasound beam encounters regions of either bone or air



Artifacts in Ultrasound Imaging cont.

- Acoustic enhancement
  - Occurs when there is an area of low attenuation relative to the surrounding tissue
  - Structures lying deeper and in-line with this area show an artificially high signal intensity
  - Areas containing a high proportion of water such as cysts can show this effect



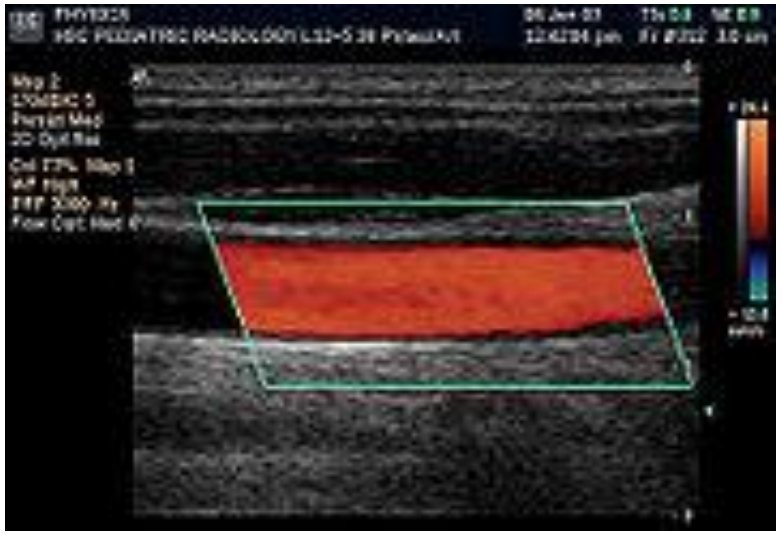
Artifacts in Ultrasound Imaging cont.

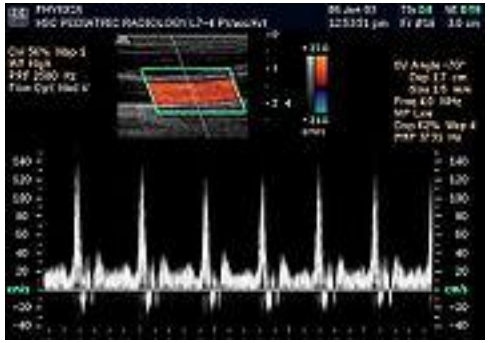
- Acoustic shadowing
  - Occurs when a highly attenuating medium results in a dark area deeper below the highly attenuating medium
  - Examples: Solid tumours, calcified structures



Doppler Ultrasound

- To measure the velocity of moving particles (usually blood)
  - Using Doppler frequency shift
  - Direction relative to the probe
    - Towards or away
- Visualization
  - Colour Doppler
  - Power Doppler





Ultrasound Imaging System

